1 Cache

Steps	Write-Through (WT)				Write-Back (WB)	
	Write-Allocate (WA)		No-Write-Allocate (NWA)		Write-Allocate (WA)	
	Fetch-on-Miss	No-Fetch-on-Miss	Write-Around	Write-Invalidate	Fetch-on-Miss	No-Fetch-on-Miss
1	Pick replacement	Pick replacement			Pick replacement	Pick replacement
2				Invalidate tag	[Write back]*	[Write back]*
3	Fetch block				Fetch block	
4	Write cache	Write partial cache			Write cache	Write partial cache
5	Write memory	Write memory	Write memory	Write memory		

- Write-once/Streaming writes = WT+NWA+WA. I/O Mapped Memory/Network Buffers/Shared Memory Context = WT+NWA+WI. Temporal/spatial write locality = WB+WA+FoM. Block aligned writes (i.e. log data) = WB+WA+NFoM.
- Index bits (idx) = $\log_2 \frac{\text{No. of entries}}{\text{Associativity}}$. Cache Tag bits = PA bits - idx - $\log_2$ (Block Size).
- Cache Optimisations: Multi-level caches & inclusion, Victim caches, Pseudo-associative caches (Search N ways sequentially), Skew-associative caches (One hash function per way), Critical Word First, Non Blocking Cache (Miss Status Handling Register), Prefetching (Simple Sequential, Stream, Strided), Multi-porting/Multi-banked

2 Coherence

- Invariants: Single-Writer Multiple Reader, Data Value (value at the start of epoch is same as value at the end of its last write epoch)
- Protocols: Valid/Invalid, Modified/Owned(Responds to GetS Request)/Exclusive/Shared/Invalid
- Protocol Design: Snooping vs Directory, Invalidate vs Update
- Locks: Spin, w/ Backoff, w/ Test, Ticket, Queuing, Load-Linked & Store Conditional
- Models: SC, TSO (Relax Write and Other Cores' Read), Relaxed/Weak Consistency
- Store Atomicity: Multiple-copy atomic (MCA), Read-Own-Write-Early MCA (rMCA) - store visible to all remote cores simultaneously, Non-multiple-copy (nMCA) - store visible to some remote cores before others

3 Memory

- Hierarchy: Channel (Independent/Lock Step - Bigger Byte in Bus), Rank (64 bit group of chips), Bank, Row, Column
- Commands: Activate (open row), Read, Write, Precharge (close row), Refresh (Activate + Precharge)
- Timings: Row Buffer Hit: CAS Latency (t_{CL}) + $t_{transfer}$. Row Buffer Miss: RAS to CAS delay (t_{RCD}) + t_{CL} + $t_{transfer}$. Row Buffer Conflict: Row Precharge Time (t_{RP}) + t_{RCD} + t_{CL} + $t_{transfer}$.
- Address Mapping: LSB Cache Line = Offset + Low Col. Sequential - MSB Row, Rank>Bank>Channel. Random -

4 Out of Order Processor

- Branch Prediction: Branch Predictor (PHT): Simple Branch History Table (Next branch same as last one), Global Correlating Predictor, Local correlating predictor, Tournament predictor. Branch Target Buffer (BTB). Return Address Stack (RSB).
- Register Renaming: Unified Register File (Free,Used/Not Ready,Used/Ready,Architectural), Architectural Map, Renaming Map
- Re-order Buffer (ROB): On exception, flush ROB and restart from PC after handler
- Dynamic Scheduling: Instruction execute as soon as its source operands are ready and FU is available
- Store Sets: Use PC to index into Store Set ID (SSID) Table, check if Last Fetched Store Table entry for SSID has been resolved
- Multithreading: Coarse-grained Multithreaded (switch thread on cache miss), Fine-grained Multithreaded (switch on stall w/ 1 cycle context switch), Simultaneous Multithreaded (multiple threads in same pipeline stage)
- Delay Slot: Instruction is executed unconditionally. If no useful instruction, a NOP must be inserted.

5 Vector, GPU & NPU

- Arithmetic Intensity: Floating-point operations per byte read
- Roofline Model: Memory Bound on left slope, Compute Bound on right constant
- Vector: Chaining - Load/Store cannot be in same convoys (chimes), Multiple Lanes, Conditional Execution
- GPU: Grid (Vectorizable Loop), Thread Block (Executed on 1 SM, assigned by Thread Block Scheduler), Warp/SIMD Thread (by SIMD Thread Scheduler), CUDA Thread/Lane. Throughput = Clock frequency x No. of SMs x No. of lanes per SM x Divergence Penalty x Instruction Issue Rate x Fraction of arithmetic instructions.
- Systolic Architecture: Multiple computations per memory access (e.g. stagger in TPU). Peak throughput = W x W x 2 (for multiply and add) x Clock Frequency.
- Quantization: Mapping high precision floating point weights and activations to lower precision representations

6 Virtualization

- Key Properties: Resource Sharing, Isolation. (Partitioning) Checkpoint/Restore, Migration, Execution Replay. (Encapsulation)
- Deployment: Hypervisor provides its own device drivers and services vs Hosted leverages host OS drivers and services
- Compatibility: Full Virtualization (virtualization transparent to guest OS and apps, performance issues) vs Paravirtualization (virtualization transparent to apps but not guest OS, does not work for proprietary OS)
- VMM Conditions: Safety (isolate guests/VMM), Equivalency (fidelity of results w/wo VMM), Efficiency (good performance)
- VMM Share Resources: Time multiplexing, Resource partitioning (requires remapping), Mediating hardware interfaces (Virtual device interfaces + device drivers)
- Popek & Goldberg 1974 Requirements: At least 2 execution modes (user & kernel), All sensitive instructions must be privileged
- Hardware Support for Virtualization: New VT Root for VMM and Non-Root for Guest eliminating ring compression
- Memory Virtualization: Page table shadowing (VMM intercepts paging operations and construct copy of page tables for guest), Extended Page Tables (Hardware page table walker but higher TLB miss cost = n × (m+1) + m, n = guest & m = host page tables)
- I/O Virtualization: Virtual device interface (Paravirtualize device interface or Single Root I/O Virtualization - single device appears as multiple devices, hypervisor not required in transfer), IOMMU (Virtual address translation for DMA, interrupt remapping and virtualization, memory protection for DMA)
- Benefits memory-intensive application (dedup and remap identical physical pages allowing guest to use more memory without swapping to disk) but not I/O intensive applications (significant performance overhead)

7 Networking

- Direct Memory Access: Separate engine to transfer blocks of data between I/O and memory without involving CPU
- NIC Interrupt Optimizations: Coalescing (process many packets before interrupt), Steering (choose which core to send interrupts)
- Remote Direct Memory Access: Allow NIC to read/write memory directly without remote CPU

8 Advanced Memory Management

- Virtual Index Cache: Homonym Problem (different virtual address on process switch - cache flush on process switch or PID tag), Synonym Problem (different virtual address may map to same physical address - don't cache shared pages)
- Virtually Indexed, Physically Tagged: Page Offset = Cache Index + Byte Offset. Cache size ≤ Page Size × Associativity
- Aliasing: HW check all sets where an alias could appear, SW page coloring
- MMU Optimisations: Multilevel TLB, Multiple page sizes (TLB: Duplicate - Requires page size tag bits, Hash/rehash, Page size predictor, Parallel lookup, Skewed TLBs), Multilevel page tables & Hashed page tables, Page Table Walkers, MMU Caches (Page Table Cache, Translation Cache, Translation Path Cache)
- Page Offset (PO) = log_2(Page Size). Virtual Page Number (VPN) = VA bits - PO. Physical Page Number (PPN) = PA bits - PO.
- TLB Tag Bits = VPN - idx. TLB Data Bits = PPN + status bits.

9 Non Volatile Storage

- Hierarchy: Controller, Channel, Chip, Block (64 - 256 pages), Page (512B - 8KB data)
- Operations: Read (20-80μs), Write (Only 1 to 0 transitions allowed, consecutive pages in block, 1-2ms), Erase block to 1 (3-7ms)
- Flash Translation Layer: Expose block based interface, equalise wear, state distributed among blocks
- Write Amplification: Number of physical pages written to flash for every logical page written by user

10 Datacenter

- Power usage effectiveness (PUE) Optimizations: Reduce number/increase efficiency of conversions, High voltage close to load, DC distribution within rack, per-rack or per-sever 12V battery. Cooling: Higher DC temperatures, Better airflow handling, Air-side economization, Water-side economization, Waste-heat energy reclamation
- Total Cost of Ownership (TCO) = Capital Expenses (CAPEX - Facilities, compute, storage, networking) + Operation Expenses (OPEX - energy, maintenance, employees)
- Mean time between failures (MTBF) = Mean time to failure (MTTF - Working) + Mean time to repair (MTTR - Non Working)
- Steady state availability = MTTF / (MTTF + MTTR)

11 Microarchitectural Attacks & Defenses

- Architectural isolation: Attacker cannot read or write victim's architectural state (indirection, resource partitioning, privilege checks)
- Hardware side-channel attacks: A transmitter (victim) instruction modulates a channel (hardware resource) in an operand-dependent manner, a receiver (attacker) observes the channel to infer the operand value. e.g. Flush+Reload
- Constant-Time programming: Mode-enabled (disables HW optimizations that can leak safe operands), Always enabled
- Transient Execution: Execution of instructions that are not architecturally committed due to prior exception or misprediction
- Meltdown attack: Faulting instruction, Spectre attack: Mispredicting instruction
- Software Defense Solution: Enable software to constrain its transient control-flow and data-flow, Eliminate (code transformations) + block (speculation fences) taint primitives (non-constant-address load, non-constant-address store, uninitialized stack load, non-argument register)